LESSON 6.15

Simplify Algebraic Expressions

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can simplify algebraic expressions by combining like terms.

WHY THIS MATTERS The music studio is placing supply orders for three rehearsal rooms. Room A needs 4 microphones and 3 stands. Room B needs 2 microphones and 5 stands. Room C needs 6 microphones and 1 stand. Instead of ordering separately, you can combine: (4 + 2 + 6) microphones and (3 + 5 + 1) stands = 12 microphones and 9 stands. That's simplifying by combining like items!

REMEMBER

Simplify Algebraic Expressions

Combining Like Terms Like terms are terms with the same variable raised to the same exponent, like 2x and 5x. To simplify an expression, you add or subtract the like terms to make it shorter.

1. Like Terms: Terms that share the exact same variable part and exponent (e.g. 4x and 2x).
2. Add or subtract the coefficients of like terms; keep the variable unchanged.
3. Constants (numbers without variables) combine together separately

Words: **Like Terms *(Términos semejantes)*** **Combine *(Combinar)*** **Simplify *(Simplificar)*** **Coefficient *(Coeficiente)*** **Term *(Término)***

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

A music festival needs supplies for 3 stages. Stage 1 needs 8m + 5 cables, Stage 2 needs 3m + 7 cables, and Stage 3 needs 4m + 2 cables, where m represents the number of microphones per stage. The total order is (8m + 5) + (3m + 7) + (4m + 2). What does the 15m tell the festival?

(A) 15 cables are needed for every microphone
(B) There are 15 microphones
(C) 15 stages need cables
(D) The cables cost $15

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Expand 5(x + 3).

(A) 5x + 15

(B) 5x + 3

(C) 8x

(D) x + 15

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which shows 6n + 18 factored using its GREATEST common factor?

(A) 2(3n + 9)

(B) 3(2n + 6)

(C) 6(n + 3)

(D) 6(n + 18)

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Expand 4(2y − 5).

(A) 2y − 20

(B) 6y − 20

(C) 8y − 20

(D) 8y − 5

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Use the distributive property to compute 7 × 62 mentally: 7 × (60 + 2) = ?

(A) 62
(B) 420
(C) 427
(D) 434

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

1. **Split** *Break one factor apart.*
2. **Multiply** *One part at a time.*
3. **Add** *Add the parts.*

6 **Circle the letter of the best answer. Show how you know.**

A music festival needs supplies for 3 stages. Stage 1 needs 8m + 5 cables, Stage 2 needs 3m + 7 cables, and Stage 3 needs 4m + 2 cables, where m represents the number of microphones per stage. The total order is (8m + 5) + (3m + 7) + (4m + 2). Simplify (8m + 5) + (3m + 7) + (4m + 2).

(A) 12m + 17
(B) 15m + 14
(C) 15m + 15
(D) 29m

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What rule determines whether two terms are 'like terms'? Why can't you combine unlike terms?

SENTENCE STARTER Two terms are like terms when they have the same ___. You cannot combine ___ and ___ because they represent different ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.15

Simplify Algebraic Expressions

Set B · Version B · Second form · core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can simplify algebraic expressions by combining like terms.

WHY THIS MATTERS The music studio is placing supply orders for three rehearsal rooms. Room A needs 4 microphones and 3 stands. Room B needs 2 microphones and 5 stands. Room C needs 6 microphones and 1 stand. Instead of ordering separately, you can combine: (4 + 2 + 6) microphones and (3 + 5 + 1) stands = 12 microphones and 9 stands. That's simplifying by combining like items!

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

The terms are scattered across the mixing board. Sort each term into its like-term group so you know what can be combined.

Groups: x-terms (have the variable x) | Constants (just numbers)

_____ 3x

_____ 5x

_____ 2x

_____ 7

_____ 4

_____ 9

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A music festival needs supplies for 3 stages. Stage 1 needs 8m + 5 cables, Stage 2 needs 3m + 7 cables, and Stage 3 needs 4m + 2 cables, where m represents the number of microphones per stage. The total order is (8m + 5) + (3m + 7) + (4m + 2). If each stage has 4 microphones (m = 4), how many cables in total?

(A) 29

(B) 60

(C) 64

(D) 74

PER 1

3 **Circle the letter of the best answer. Show how you know.**

The board shows 3x + 5x on one channel. Simplify it.

(A) 8x

(B) $8x^2$

(C) 15x

(D) 35x

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Simplify the track: 4a + 2 + a.

(A) 5a + 2

(B) $5a^2 + 2$

(C) 6a

(D) 7a

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Clean up this channel: 7y − 3y.

(A) 4y
(B) 4
(C) 10y
(D) 21y

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Simplify the mix: 6m + 4 + 3m.

(A) 9m + 4
(B) 9m + 4m
(C) 10m
(D) 13m

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What rule determines whether two terms are 'like terms'? Why can't you combine unlike terms?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help